# Week 6 Notes

STAT 251 SECTION 01

This week has three parts. On **Monday** we finish the tools for describing a single distribution: the normal distribution, $z$-scores, and what they let us say about outliers. On **Wednesday** we use those tools on a new kind of distribution, the distribution of a *statistic*, which is the bridge from describing data to drawing conclusions from it. On **Friday** we start probability, the language that makes that bridge precise.

# The normal distribution

## The data

Two real data sets run through the first half of this week.

**Female student heights.** A survey of 379 college students recorded each student's height and gender; 262 of them are women.

- $X$ = the height of a female student, in inches

| HEIGHT | GENDER |
|---|---|
| 56 | Female |
| 57 | Female |
| 58 | Female |
| 60 | Female |
| 60 | Female |
| ... | ... |

The file is sorted from shortest to tallest, which is why the first rows are all short. $n = 262$.

**Central Park in January.** The National Weather Service has recorded the mean temperature in Central Park, New York, for every month since 1869.

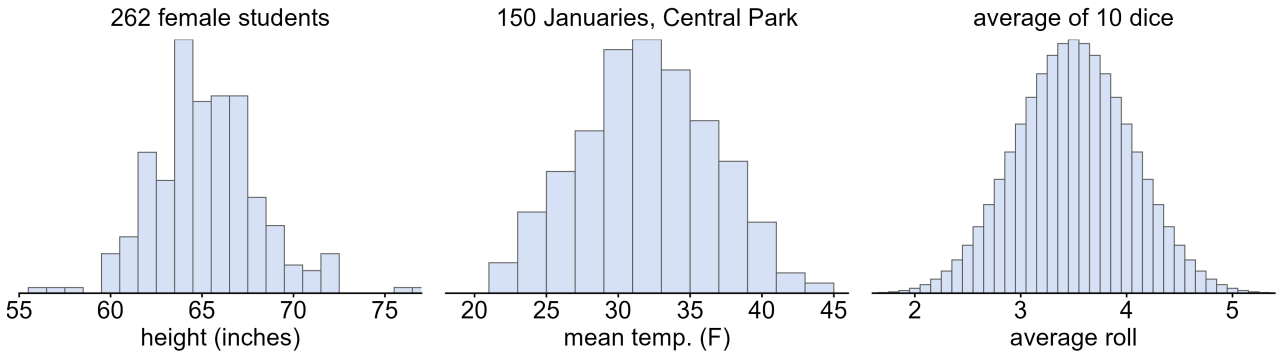
- $X$ = the mean January temperature in Central Park in a given year, in °F

| YEAR | JAN | FEB | MAR | DEC | ANNUAL |
|---|---|---|---|---|---|
| 1869 | 35.1 | 34.5 | 34.8 | 34.7 | 51.4 |
| 1870 | 37.5 | 31.3 | 34.1 | 34.1 | 53.6 |
| 1871 | 28.3 | 30.2 | 44.2 | 29.2 | 51.1 |
| 1872 | 28.8 | 29.9 | 30.5 | 26.7 | 51 |
| 1873 | 28.6 | 29.5 | 35.7 | 36.5 | 51 |
| ... | ... | ... | ... | ... | ... |

$n = 150$ years, 1869 to 2018. We use only the January column.

## What do these have in common?

Here are three distributions that have nothing to do with each other: the heights of the 262 female students in our running data set, the mean January temperature in Central Park, New York, in each of the 150 years from 1869 to 2018, and the average of ten rolls of a fair die.



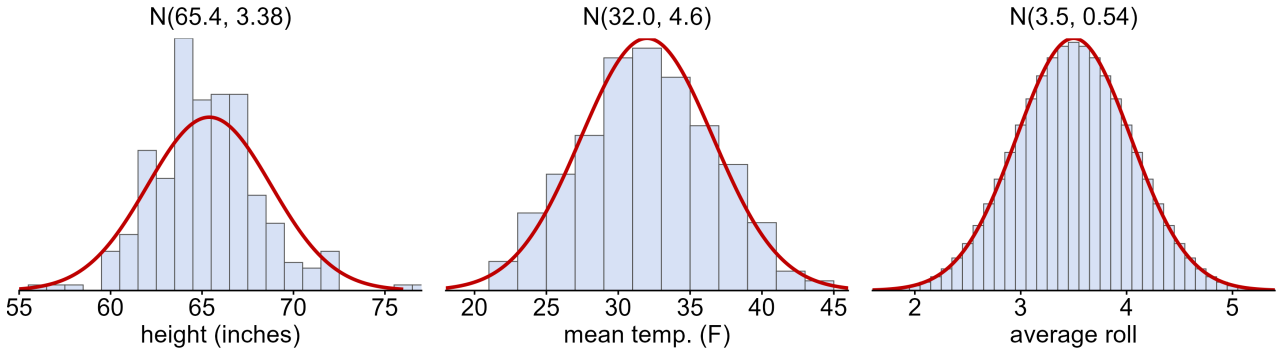

- What do the three have in common?

Each one is single-peaked, symmetric and bell shaped: most values pile up near the middle, and they thin out at the same rate on both sides. This shape turns up so often that it has a name.

## The normal distribution

- A **normal distribution** is a symmetric, bell-shaped curve with its peak at the mean.

Below, the same three distributions each have a normal curve drawn over them. It is the same *shape* each time; only its center and its spread change.




The heights follow the curve least closely: they are lumpier, and a few students sit well out in the tails. No real data set is exactly normal. The normal curve is a **model**, and the question is always whether it is close enough to be useful.

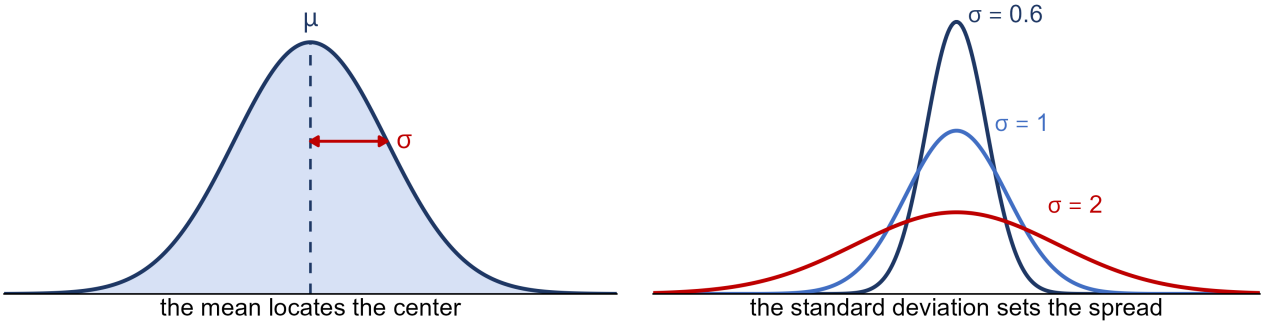
## Two parameters: $\mu$ and $\sigma$

Every normal curve is pinned down by just two numbers, its **parameters**. We write

$$X \sim N(\mu, \sigma)$$

for a variable that is normally distributed with mean $\mu$ and standard deviation $\sigma$. The titles on the curves above are exactly this: the heights are roughly $N(65.4, 3.38)$ and the January temperatures roughly $N(32.0, 4.6)$.

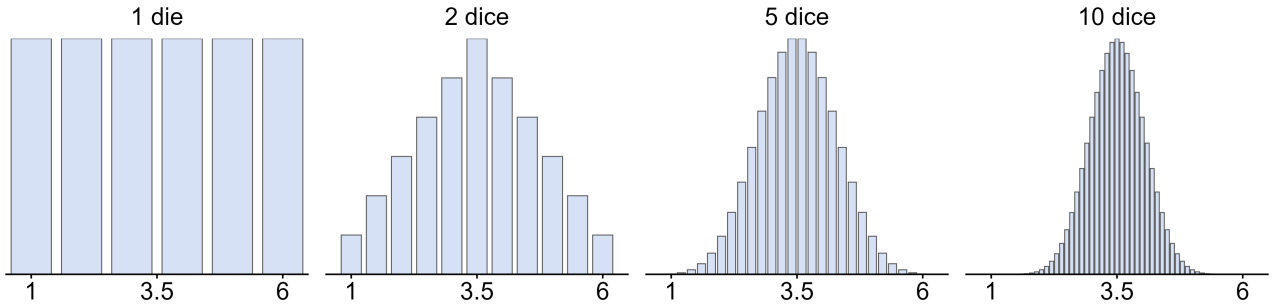
- The mean $\mu$ **locates the center**. Changing $\mu$ slides the curve along the axis without changing its shape.
- The standard deviation $\sigma$ **sets the spread**. A small $\sigma$ gives a tall, narrow curve; a large $\sigma$ gives a short, broad one. The center does not move.




Greek letters are for the population ($\mu$, $\sigma$) and Roman letters for a sample ($\bar{x}$, $s$), the same convention as parameters and statistics in the Weeks 4-5 notes.

## Why is the normal so important?

Why would one curve matter so much? Look at what happens when we average dice. One die is flat: every face is equally likely. The average of two dice is already peaked in the middle, and the average of five or ten is a bell.

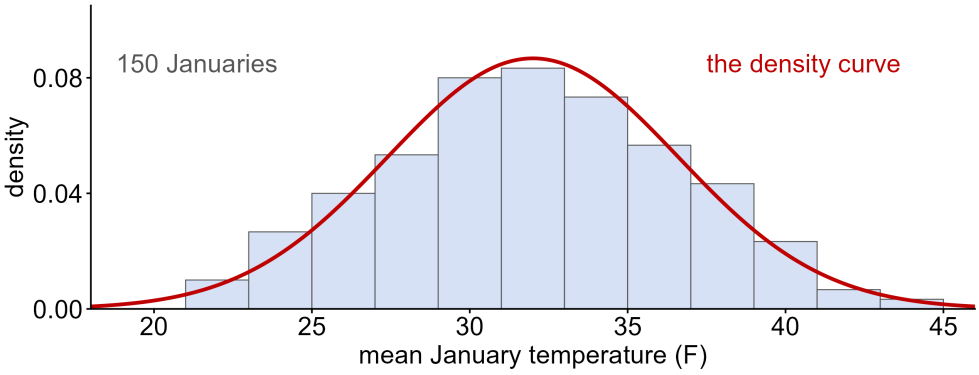



- **Many measurements are approximately normal**, especially physical ones such as heights, and quantities such as a month's average temperature that are themselves averages of many days.
- **Averages of random values become normal, whatever the individual values look like.** The dice above start out flat, not bell shaped at all. This is the central limit theorem, and it is why a *sample mean* is approximately normal. That is Wednesday's topic.
- **Most of the methods in the rest of this course are built on it**: margins of error, confidence intervals and hypothesis tests all lean on the normal curve.

## Density curves

What exactly *is* the smooth curve we keep drawing over histograms? A histogram's bars are counts inside intervals that *we* chose. A continuous variable, such as a temperature, has no natural intervals, so in theory its distribution is better drawn as a smooth curve.

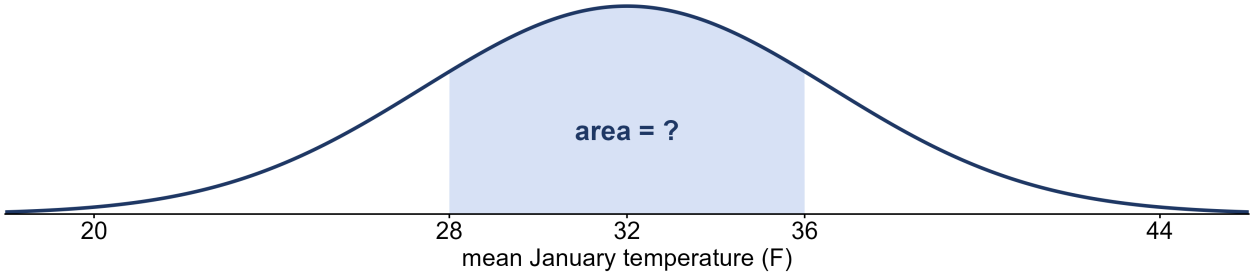
- A **density curve** is a smooth curve that describes a distribution. It never dips below the axis, and the total area under it is exactly 1.



The bars are the 150 Januaries we actually recorded. The curve is the distribution behind them, a model of what January in Central Park looks like in the long run. The vertical axis is labelled **density**, and it is worth getting a rough feel for what that means.

## Area under the curve is proportion

The January temperatures have $\bar{x} = 32.0$°F and $s = 4.6$°F. What fraction of Januaries averaged between 28°F and 36°F?




On a density curve, **the area under the curve over an interval is the proportion of values in that interval**. The shaded area works out to 0.62. In the actual data, 92 of the 150 Januaries fall in that range, a proportion of 0.61. The model and the data agree closely.

That gives a loose working idea of **probability density**:

- The **height** of the curve is the density. It shows where values crowd together: the curve is tallest near 32°F because that is where Januaries are most common. The height on its own is *not* a proportion or a probability.
- The **area** under the curve is the proportion. If we pick one January at random, the area over an interval is the *probability* that it lands in that interval. That is why the total area has to be 1: every January lands somewhere.

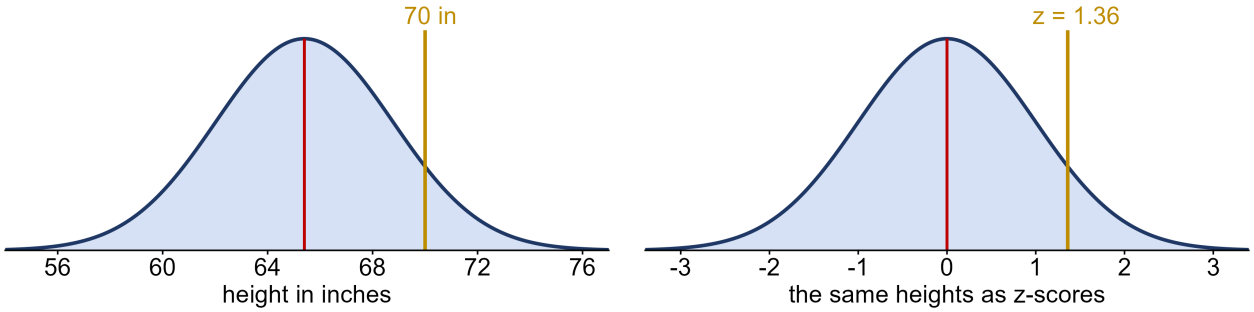
We will make this precise when we reach probability on Friday and continuous random variables in Week 7. For now: **height means crowded, area means how many**. Everything else about the normal distribution this week, including the empirical rule, the $z$-table and `pnorm`, is a way of finding areas under this curve.

## The standard normal distribution

There are infinitely many normal curves, one for every choice of $\mu$ and $\sigma$. One of them gets a name of its own.

- The **standard normal distribution** is the normal distribution with mean 0 and standard deviation 1, written $Z \sim N(0, 1)$.

Every normal distribution is a shifted and stretched copy of this one. If we measure how far each value is from $\mu$ **in units of** $\sigma$, any normal curve turns into $N(0, 1)$. Below, the heights of the 262 female students are drawn twice: once in inches, and once as the number of standard deviations each height is from the mean.




It is the same curve; only the labels on the axis change. That "number of standard deviations from the mean" is called a $z$-score, and it is the next topic.

# $z$-scores

## The data: breakfast cereals

The twenty US breakfast cereals from Week 3, with their sodium and sugar content.

- $Y$ = the sodium content of a breakfast cereal, in mg

| Cereal | Sodium | Sugar | Type |
|---|---|---|---|
| Frosted Mini Wheats | 0 | 11 | A |
| Raisin Bran | 340 | 18 | A |
| All Bran | 70 | 5 | A |
| Apple Jacks | 140 | 14 | C |
| Captain Crunch | 200 | 12 | C |
| ... | ... | ... | ... |

$n = 20$ cereals.

## Which observation is more extreme?

How do we say how far a value sits from the center of its distribution? So far, "far from the center" has been measured in the units of whatever we happened to be measuring. That makes it impossible to compare across variables. Consider two samples, each with $n = 20$ observations:

- $X$ = an exam score (points): $\bar{x} = 74$, $s = 8$, and one student scored $x = 90$.
- $Y$ = the sodium in a breakfast cereal (mg): $\bar{y} = 167$, $s = 77$, and one cereal has $y = 290$.

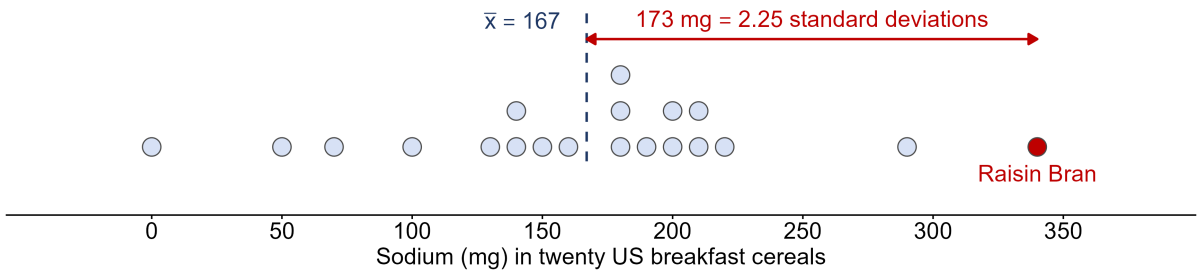
Which observation is more extreme? The cereal is $123$ mg above its mean and the student only $16$ points above theirs, but points and milligrams are different scales, so the two raw deviations can't be compared. The fix is to measure each deviation in **standard deviations of its own variable**:

$$\text{student: } \frac{90 - 74}{8} = \frac{16}{8} = 2.0 \qquad \text{cereal: } \frac{290 - 167}{77} = \frac{123}{77} \approx 1.6$$

The student is 2.0 standard deviations above the mean and the cereal only 1.6, so the exam score is the more extreme of the two, even though its raw deviation is far smaller. On a normal curve this is the natural yardstick, because $\sigma$ is exactly what sets the curve's spread.

## Measuring a deviation in standard deviations

Recall the twenty US breakfast cereals from the Week 3 notes, with mean sodium $\bar{x} = 167$ mg and standard deviation $s \approx 77$ mg. The cereal furthest above the mean is Raisin Bran, at 340 mg.




Raisin Bran is $340 - 167 = 173$ mg above the mean, which is $173/77 \approx 2.25$ standard deviations. Dividing a deviation by the standard deviation strips the units off, and what is left means the same thing no matter what was measured. That number is the **$z$-score**:

$$z = \frac{x - \bar{x}}{s} = \frac{\text{observation} - \text{mean}}{\text{standard deviation}}$$

A positive $z$ is above the mean and a negative $z$ is below it. A cereal at $z = 2.25$ and a student at $z = 2.25$ are equally unusual within their own distributions, even though one is milligrams and the other is exam points.

For a population the $z$-score uses the parameters, $z = (x - \mu)/\sigma$; for a sample it uses the statistics, $z = (x - \bar{x})/s$. To go the other way, $x = \bar{x} + z\,s$. Converting every value of a variable to a $z$-score is called **standardizing**, and standardizing a normal variable gives the standard normal distribution, $Z \sim N(0, 1)$.

### Example: a 70-inch student

Our running data set this week is the heights of the 262 female students in `heights.csv`, with $\bar{x} = 65.4$ inches and $s = 3.38$ inches. A student who is 70 inches tall has

$$z = \frac{70 - 65.4}{3.38} = 1.36$$

She is 1.36 standard deviations above the mean: the gold line on the standard normal picture above. Standardizing relabels the axis. It does not move a single student.

## Examples

### Example: Jerome's points

This one is Try It 6.2 from the textbook. Jerome averages 16 points a game with a standard deviation of 4 points, so $X \sim N(16, 4)$. One night he scores 10 points. Fill in the blanks: $x = 10$ is ___ standard deviations to the ___ (right or left) of the mean ___.

$$z = \frac{10 - 16}{4} = -1.5$$

so $x = 10$ is **1.5** standard deviations to the **left** of the mean, **16**. A negative $z$-score always means below the mean.

### Example: carbon emissions across the EU

The table below gives carbon dioxide emissions per person (metric tons) for the 28 member states of the European Union, from the World Bank.

- $X$ = the carbon dioxide emissions per person in an EU country, in metric tons ($n = 28$)

| Country | Code | CO2 per person (metric tons) |
|---|---|---|
| Austria | AUT | 8.0 |
| Belgium | BEL | 10.0 |
| Bulgaria | BGR | 6.0 |
| Croatia | HRV | 4.7 |
| Cyprus | CYP | 7.0 |
| Czech Republic | CZE | 10.7 |
| Denmark | DNK | 8.3 |
| Estonia | EST | 13.7 |
| Finland | FIN | 11.5 |
| France | FRA | 5.6 |
| Germany | DEU | 9.1 |
| Greece | GRC | 7.7 |
| Hungary | HUN | 5.1 |
| Ireland | IRL | 8.8 |
| Italy | ITA | 6.7 |
| Latvia | LVA | 3.6 |
| Lithuania | LTU | 4.4 |
| Luxembourg | LUX | 21.4 |
| Malta | MLT | 6.2 |
| Netherlands | NLD | 11.0 |
| Poland | POL | 8.3 |
| Portugal | PRT | 5.0 |
| Romania | ROM | 3.9 |

| Country | Code | CO2 per person (metric tons) |
|---|---|---|
| Slovak Republic | SVK | 6.7 |
| Slovenia | SVN | 7.5 |
| Spain | ESP | 5.8 |
| Sweden | SWE | 5.6 |
| United Kingdom | GBR | 7.9 |

The mean is $\bar{x} \approx 7.9$ and the standard deviation $s \approx 3.6$ metric tons.

- How many standard deviations above the mean is Luxembourg, at 21.4 tons per person?

$$z = \frac{21.4 - 7.9}{3.6} = 3.75$$

- How high would a country's emissions have to be to sit 2 standard deviations above the mean?

$$\bar{x} + 2s = 7.9 + 2(3.6) = 15.1 \text{ metric tons}$$

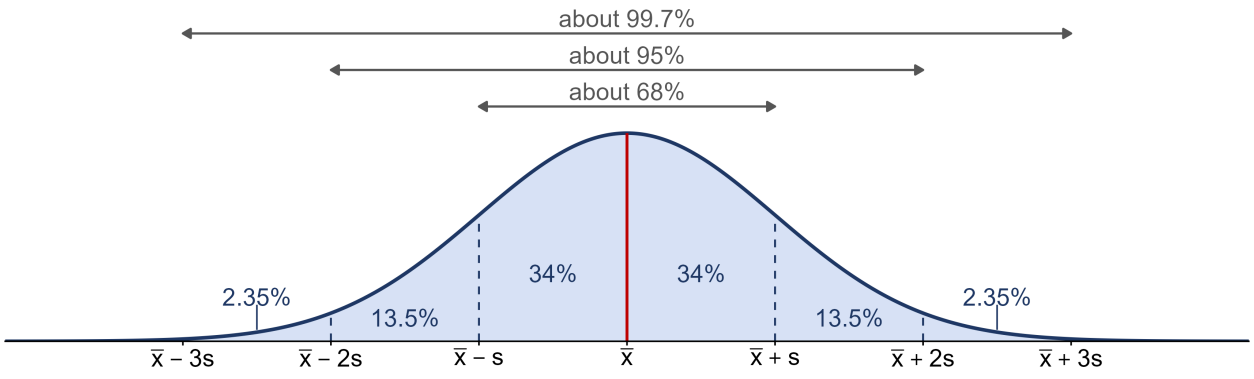
  Only Luxembourg clears that line; the next highest, Estonia, emits 13.7.

# The empirical rule and outliers

## The empirical rule

Every normal curve has the same proportions once distance is measured in standard deviations.

- The **empirical rule** (the 68-95-99.7 rule): in a bell-shaped distribution, about **68%** of the values fall within 1 standard deviation of the mean, about **95%** within 2, and about **99.7%** within 3.

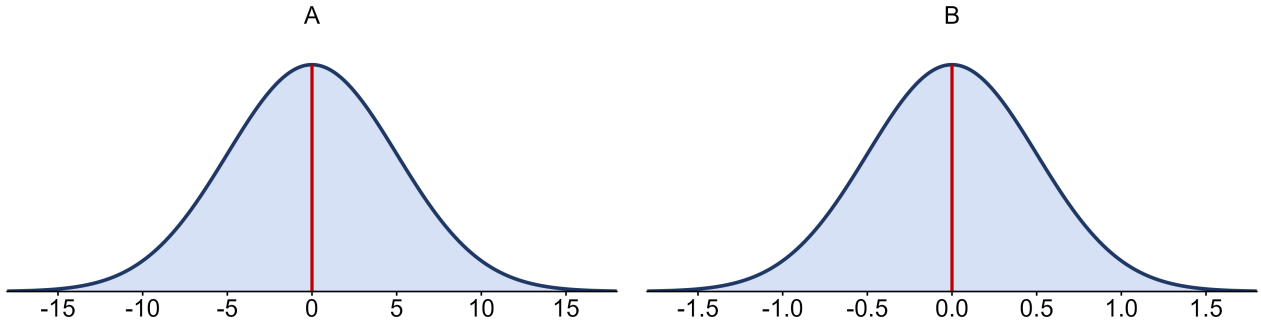


Because the curve is symmetric, the rule splits into pieces: 34% between the mean and one standard deviation on each side, 13.5% between one and two, and 2.35% between two and three, leaving 0.15% beyond three on each side. Putting pieces together answers most questions of this kind: for example $50\% + 34\% = 84\%$ of values are below $\bar{x} + s$, and $50\% + 34\% + 13.5\% = 97.5\%$ are below $\bar{x} + 2s$.

The rule is **only** for bell-shaped distributions. Applied to a skewed distribution it gives the wrong answers.

## Practice: read $\sigma$ off a curve

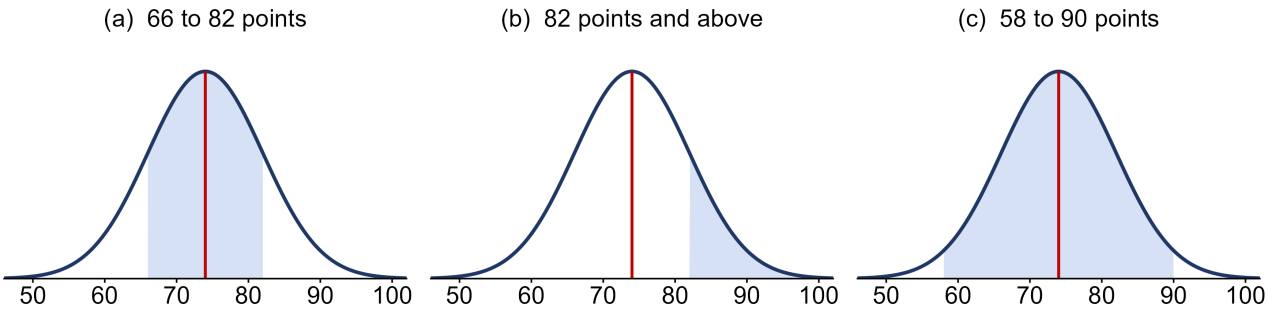
**Practice:** estimate the standard deviation of each curve.




*A: $\sigma \approx 5$. B: $\sigma \approx 0.5$. Step out from the center until you have captured roughly the middle two thirds of the curve (68%) and read that distance off the axis. The two pictures are the same shape, so the shape alone tells you nothing about the spread: always check the axis.*

## Example: back to that exam

Exam scores in a class are roughly bell shaped with $\bar{x} = 74$ and $s = 8$ points. What percentage of scores falls in each shaded region?



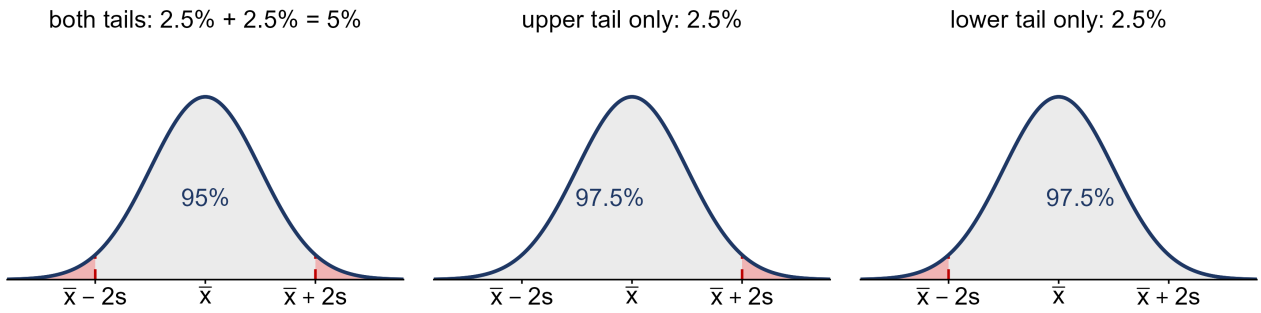

- a. 66 to 82 is $\bar{x} \pm 1s$, so about **68%**.
- b. 82 and above: 68% lie within $\pm 1s$, leaving 32% split evenly between the two tails, so about **16%**.
- c. 58 to 90 is $\bar{x} \pm 2s$, so about **95%**.

Part (b) used the empirical rule and symmetry together. Notice that the student who scored 90 in "Which observation is more extreme?" is exactly 2 standard deviations up: only about 2.5% of the class did better.

## Identifying outliers: the two-standard-deviation rule

The empirical rule gives a natural definition of "unusual" for bell-shaped data.

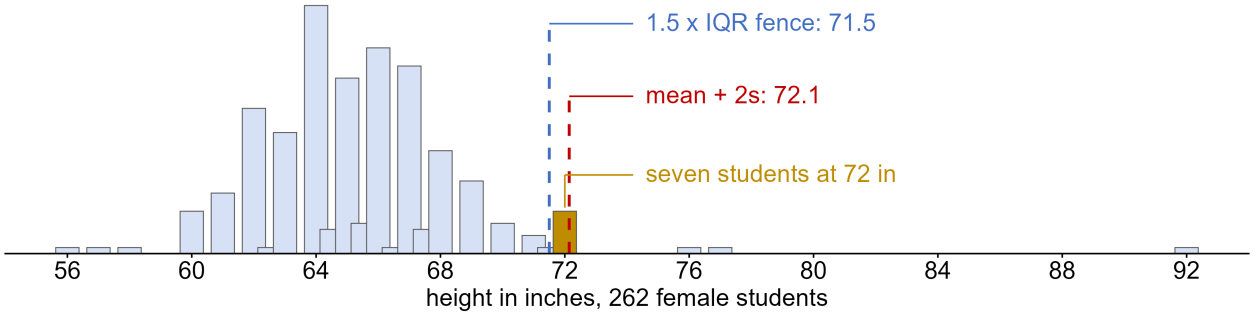
- An **outlier** under the $\pm 2s$ rule is a value at least 2 standard deviations from the mean, $|z| \geq 2$. Only about 5% of the values of a bell-shaped distribution are that far out. A stricter version uses 3 standard deviations, about 0.3%.




Ask first whether you care about one direction or both: about 2.5% of values lie beyond $+2s$ alone, and 5% beyond $\pm 2s$ together. By this rule Luxembourg ($z = 3.75$) is an outlier, and a large one: beyond 3 standard deviations is only about 0.15% of a bell-shaped distribution in each tail.

### When the two outlier rules disagree

You already know the $1.5 \times IQR$ rule from Week 3. Do the two rules flag the same students?




Not here. The IQR rule's upper fence is $Q_3 + 1.5 \times IQR = 71.5$ inches and it flags 13 of the 262 women; the $\pm 2s$ fence is $\bar{x} + 2s = 72.1$ inches and it flags only 6. The seven students at 72 inches are outliers under one rule and ordinary under the other. The single 92-inch value (almost certainly a data-entry error) drags the mean up and inflates $s$, which pushes the $2s$ fence out past the 72s. The quartiles barely notice one extreme value; the mean and standard deviation always do.

Neither rule is “the truth”. An outlier is a value worth a second look, not a value to delete.

# Finding areas, and transformations

## Computing quantiles from a normal distribution

Our 70-inch student has $z = 1.36$. What percentage of the women are shorter than she is? If her $z$-score were exactly 1, the empirical rule would give $50\% + 34\% = 84\%$. But 1.36 is not 1, 2 or 3, and the empirical rule only knows whole standard deviations. For anything in between we need the actual area under the curve to the left of $z$.

Because every normal distribution has the same areas once converted to $z$, one table covers all of them. For decades that table was how it was done:

- The **standard normal table** ($z$-table) gives, for each $z$, the area under the standard normal curve to the **left** of $z$. Find the row for the first two digits of $z$ and the column for the third.

| z | .00 | .01 | .02 | .03 | .04 | .05 | .06 | .07 | .08 | .09 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1.0 | 0.8413 | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554 | 0.8577 | 0.8599 | 0.8621 |
| 1.1 | 0.8643 | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770 | 0.8790 | 0.8810 | 0.8830 |
| 1.2 | 0.8849 | 0.8869 | 0.8888 | 0.8907 | 0.8925 | 0.8944 | 0.8962 | 0.8980 | 0.8997 | 0.9015 |
| 1.3 | 0.9032 | 0.9049 | 0.9066 | 0.9082 | 0.9099 | 0.9115 | **0.9131** | 0.9147 | 0.9162 | 0.9177 |
| 1.4 | 0.9192 | 0.9207 | 0.9222 | 0.9236 | 0.9251 | 0.9265 | 0.9279 | 0.9292 | 0.9306 | 0.9319 |
| 1.5 | 0.9332 | 0.9345 | 0.9357 | 0.9370 | 0.9382 | 0.9394 | 0.9406 | 0.9418 | 0.9429 | 0.9441 |
| 1.6 | 0.9452 | 0.9463 | 0.9474 | 0.9484 | 0.9495 | 0.9505 | 0.9515 | 0.9525 | 0.9535 | 0.9545 |

Row 1.3, column .06 gives $0.9131$, so about 91% of the women are shorter than 70 inches. (In the actual data it is 243 of 262, or 92.7%, which is close: the heights are only approximately normal.) The area to the **right** of $z$ is one minus the table value, here $1 - 0.9131 = 0.0869$.

Today software does the lookup. In R:

- **From a value to an area:** `pnorm(1.36)` returns 0.9131, the same number as the table. `pnorm(70, mean = 65.4, sd = 3.38)` skips the $z$-score step entirely.
- **From an area to a value:** `qnorm(0.90)` returns 1.28, so 90% of any normal distribution lies below $\bar{x} + 1.28s$. For our heights that is $65.4 + 1.28(3.38) = 69.7$ inches.

The normal calculator on the course tools site (Resources → Tools/Calculators) does both. Either way, you still have to know which number you are asking for.

## A note about transformations of variables

We often need to change the units of measurement of a variable, such as from Fahrenheit to Celsius, feet to meters, or dollars to euros. Changing a variable like this is called a **transformation**, and transformations come in two kinds.

- **Linear transformations** are made of adding, subtracting, multiplying and dividing. They take the form

  $$y = ax + b$$

  where $a$ is a *scaling* constant, $b$ is a *shifting* constant, $x$ is the original variable and $y$ is the transformed variable. Inches to centimeters is $a = 2.54$, $b = 0$.
  - The $z$-score is a linear transformation, with $a = 1/s$ and $b = -\bar{x}/s$.
  - Linear transformations **preserve the shape** of a variable's distribution.
- **Nonlinear transformations** include squaring, taking roots, logarithms and exponentiation.
  - They do **not** preserve the shape of a variable's distribution, which is often exactly why they are used: a logarithm, for example, can pull in a long right tail.

What a linear transformation does to the summaries:

- **Center** moves with the whole transformation: $\bar{y} = a\bar{x} + b$, and the median behaves the same way.
- **Spread** ignores the shift: $s_y = |a|\, s_x$ and $IQR_y = |a|\, IQR_x$. Adding $b$ slides every value by the same amount, so the distances between them do not change.
- **Shape** does not change at all. Skewness, gaps, clusters and outliers all survive exactly as they were.

## Example: inches to centimeters

Converting our 262 heights to centimeters multiplies every value by 2.54, so $a = 2.54$ and $b = 0$.

- The mean becomes $\bar{y} = 2.54(65.4) = 166.1$ cm and the standard deviation $s_y = 2.54(3.38) = 8.59$ cm.
- The 70-inch student is $2.54(70) = 177.8$ cm tall, and her $z$-score is

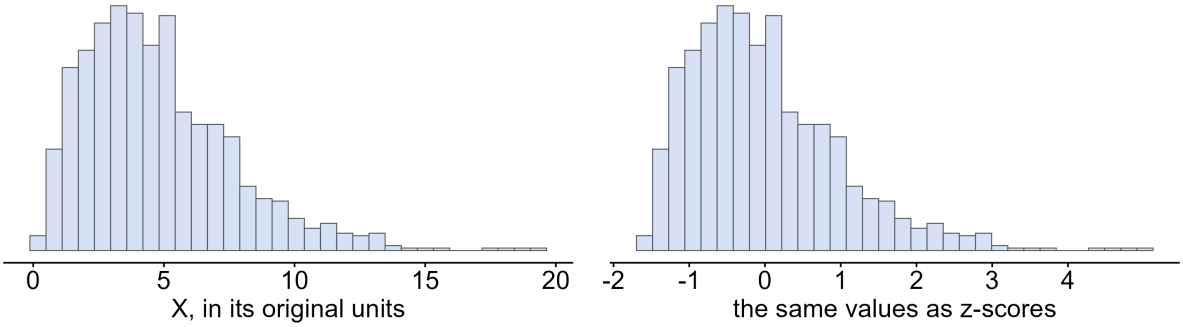
  $$z = \frac{177.8 - 166.1}{8.59} = 1.36$$

  exactly what it was in inches.

Both the deviation and the standard deviation were multiplied by 2.54, and the factor cancels. A $z$-score has no units, so it cannot depend on which units you chose. That is the whole reason a $z$-score lets us compare an exam score with a breakfast cereal.

### A $z$-score does not make data normal

A common mistake is to think that standardizing a skewed variable straightens it out. It does not: standardizing is a linear transformation, so it moves the center to 0 and sets the spread to 1 and leaves the shape exactly as it found it.




Both pictures show the same 1,000 values; the right-hand axis just reads in standard deviations. It also means the empirical rule and the $\pm 2s$ outlier rule still do not apply to the picture on the right.

# Sampling distributions

On Monday every distribution was a distribution of *data*: exam scores, sodium, heights. Today we look at a distribution of a *statistic*, and it turns out to be the most important distribution in the course.

## Statistics vary from sample to sample

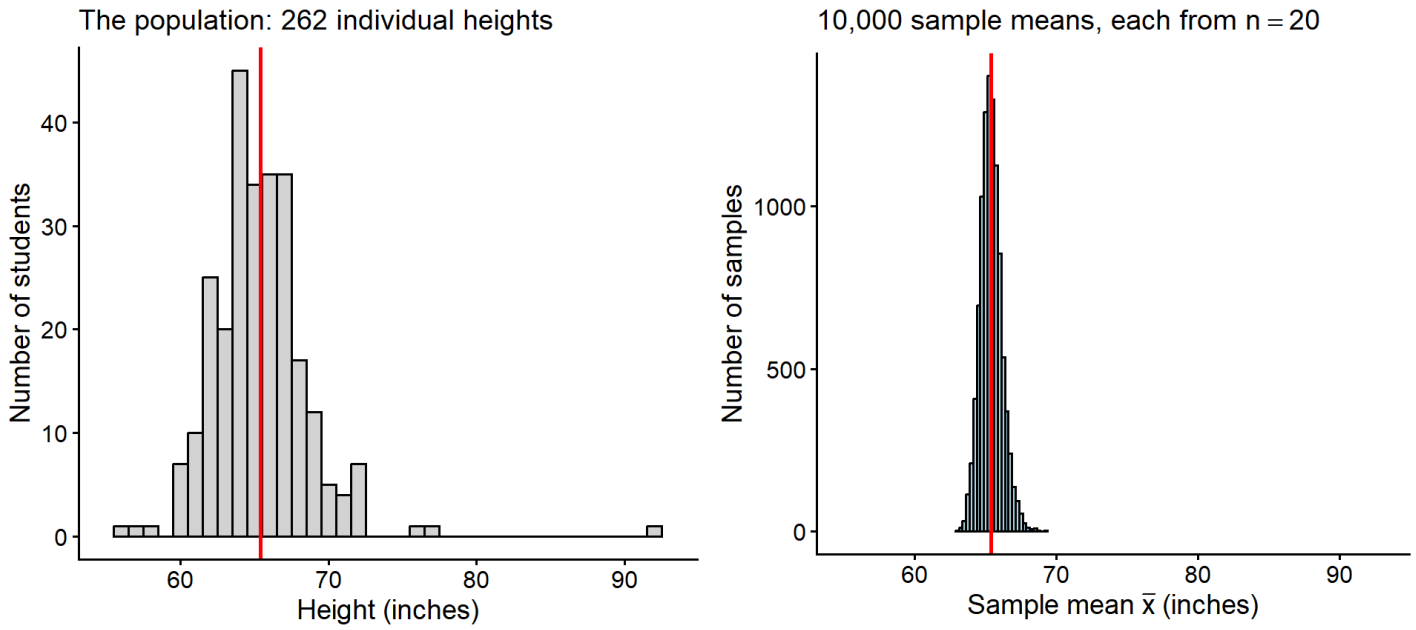
Recall the distinction from the Weeks 4-5 notes. A **parameter** is a number that describes the population, such as its mean $\mu$; a **statistic** is a number computed from a sample, such as $\bar{x}$. We almost never know the parameter, so we use the statistic to estimate it.

To see what happens, let's pretend for a moment that we *do* know the population. Treat the 262 female students as the whole population, with $\mu = 65.4$ inches and $\sigma = 3.38$ inches, and take a random sample of $n = 20$ of them. The first sample has $\bar{x}_1 = 66.33$ inches. A second random sample of 20 gives $\bar{x}_2 = 64.85$ inches.

Neither sample mean equals $\mu$, and they don't equal each other. Nothing went wrong: a different sample contains different people, so it gives a different answer. This is the **sampling error** from the Weeks 4-5 notes, the difference between a statistic and the parameter it estimates, caused purely by which units happened to be chosen.

## The sampling distribution

Now imagine repeating the process over and over: draw a sample of 20, compute $\bar{x}$, write it down, and start again. Below are the means of 10,000 such samples.




- The **sampling distribution** of a statistic is the distribution of its values over all possible samples of the same size $n$ from the same population.

The name is new, but the idea is not. In Week 3 we saw that a sample mean moves from one sample to the next (**sampling variation**) and measured that movement with the **standard error**, $SE = s/\sqrt{n}$: the standard error is simply the standard deviation of the sampling distribution. And the histograms in "A bigger sample does not fix bias" in the Weeks 4-5 notes, 5,000 simple random samples against 5,000 Rec Center surveys, *were* sampling distributions: each bar counted how many of the repeated surveys gave a particular estimate.

The two histograms are drawn on the same axis on purpose. Compare them:

- **Center.** The sample means are centered at 65.39 inches, right on $\mu$ (the red line). On average $\bar{x}$ neither overshoots nor undershoots, so it is **unbiased**.
- **Spread.** The sample means are far less spread out than the individual heights. One student can easily be 72 inches tall, but a sample of 20 students whose *average* is 72 inches essentially never happens, because the tall and short students in a sample cancel each other out.
- **Shape.** The individual heights are lumpy, with a stray value at 92 inches. The sample means form a smooth, symmetric bell. That is not luck: it is the central limit theorem, which we meet in

**A note on the center.** A sampling distribution is centered at the population parameter **only if the statistic is unbiased**. A statistic is **unbiased** when the mean of its sampling distribution equals the parameter it estimates. The sample mean $\bar{x}$ from a random sample is unbiased, which is why the histogram above sits right on $\mu$. It is not automatic:

- **a biased design** moves the whole sampling distribution. The Rec Center surveys in the Weeks 4-5 notes piled up around 75% when the truth was 40%, however many times they were repeated;
- **some statistics are biased even with random sampling.** Computing the sample variance by dividing by $n$ instead of $n-1$ (Week 3) gives a sampling distribution centered slightly *below* $\sigma^2$.

Bias moves the center of the sampling distribution; the standard error describes its spread.

## Standard error

The spread of a sampling distribution gets its own name, because it is exactly the "how far off might my estimate be" that we care about.

- The **standard error** (SE) of a statistic is the standard deviation of its sampling distribution. For a sample mean it is

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

  and when $\sigma$ is unknown we estimate it with $s/\sqrt{n}$.

For our samples of 20, $SE = 3.38/\sqrt{20} = 0.755$ inches. The standard deviation of the 10,000 simulated means is 0.756, so the formula matches what actually happened.

Keep the two standard deviations straight. $\sigma = 3.38$ describes how far a single student's height typically falls from $\mu$. $SE = 0.76$ describes how far a whole sample's *average* typically falls from $\mu$.

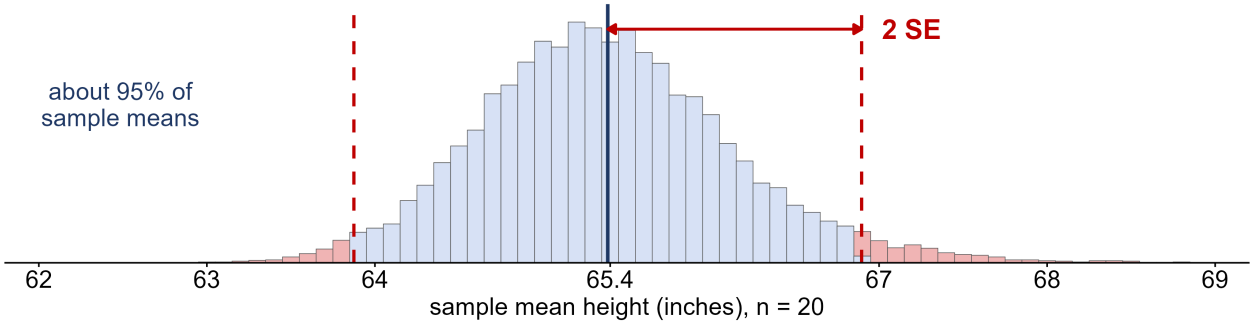
The $\sqrt{n}$ in the denominator means a bigger sample gives a smaller standard error, but slowly. With $n = 80$, four times as many students, $SE = 3.38/\sqrt{80} = 0.378$ (simulated: 0.377): quadrupling the sample only halves the error. And remember from the Weeks 4-5 notes that a bigger sample shrinks the standard error but does nothing about bias.

## Margin of error

Because the sampling distribution of $\bar{x}$ is bell shaped, the empirical rule applies to it. About 95% of sample means land within 2 standard errors of $\mu$. In the simulation, 95.1% of the 10,000 sample means did.

- The **margin of error** of an estimate measures how far we expect the estimate to fall from the true value of the parameter. It is a measure of the *between-sample* variability of the estimate: the largest distance between the parameter and an estimate that is **not an outlier**. Using the $\pm 2 \cdot SE$ rule for outliers, it is

$$\text{margin of error} = 2 \cdot SE$$




For samples of 20 heights the margin of error is $2(0.755) = 1.51$ inches: about 95% of samples give an $\bar{x}$ within $65.4 \pm 1.5$ inches. A sample mean further than that from $\mu$ would be an outlier in the sampling distribution, and therefore rare.

Don't confuse the two errors. The **sampling error** is how far *this particular* estimate happens to be from the parameter, and we never know it, because we don't know the parameter. The **margin of error** is a bound on how large the sampling error is likely to be, and we *can* compute it before we have even taken the sample. When a poll reports "52% support, with a margin of error of 3 percentage points", this is the number they mean.

## Statistical significance

The margin of error also tells us when a result is too big to blame on chance.

- A result is **statistically significant** when it is too large to be explained by ordinary sampling variation, that is, when it would be unusual if chance alone were at work. By the $\pm 2 \cdot SE$ rule,

that means further than 2 standard errors from what we would otherwise expect: a statistically significant result is one that **falls outside the margin of error**.

Suppose someone hands you a random sample of 20 students with a mean height of 70.33 inches and claims it came from our population of women. If it did, $\bar{x}$ should almost certainly be within $65.4 \pm 1.5$ inches. Instead it is

$$z = \frac{70.33 - 65.4}{0.755} = 6.5$$

standard errors away. Ordinary sampling variation essentially never produces that, so the result is statistically significant: this sample almost certainly did not come from the female students. (It didn't. It is a random sample of the 117 male students in the same file.)

A sample mean of 66.0 inches, on the other hand, is $(66.0 - 65.4)/0.755 = 0.8$ standard errors from $\mu$. That is well within the range of ordinary variation, so it is **not** statistically significant. It could easily be a sample of our women.

Two cautions that will come up again and again:

- **Significant does not mean important.** It only means "not ordinary chance variation". With a large enough sample, a difference far too small to matter can be statistically significant.
- **Not significant does not mean no difference.** It means the data can't rule chance out, which can simply be a sign that the sample was too small.

This week's version of the idea is informal. From Week 8 on, confidence intervals and hypothesis tests make it precise, but the logic never changes: compare what you saw with how much variation chance alone would produce.

# Introduction to probability

## Why probability?

Look again at two statements from this week. On Wednesday: *about 95% of samples land within 2 standard errors of $\mu$*. And in the significance example: *a sample that far out would be very unlikely to happen by chance*. But we only ever take **one** sample. So what do "95%" and "unlikely" actually mean?

They describe what would happen if we repeated the random sampling many, many times. Statements about the long run of a chance process are statements of **probability**, and statistics needs a precise language for them.

# Quantifying uncertainty

- **Randomization.** We use chance on purpose, by randomly sampling subjects from a population in a survey or randomly assigning subjects to treatments in an experiment, to avoid bias.
- **The price: uncertainty.** By chance, the statistics from a sample are not exactly the parameters of the population.
- **The payoff.** Because chance chose the sample, the results follow the laws of mathematical probability, so we can measure how far off we are likely to be.

# The language of probability

- A **random trial** is a process or experiment with a set of well-defined possible outcomes. What makes it *random* is that more than one outcome is possible: rolling a die, flipping a coin, or choosing a simple random sample.
- The **sample space** $S$ is the collection of all possible outcomes. For one roll of a six-sided die, $S = \{1, 2, 3, 4, 5, 6\}$.
- An **event** is any possible outcome of a random trial, written with a capital letter. For example $A$ = roll an even number $= \{2, 4, 6\}$.
- The **probability** of an event $A$, written $P(A)$, is a number between 0 and 1 that measures how likely $A$ is: the proportion of times $A$ occurs in a very long series of repetitions of the random trial. A probability of 0 means $A$ never happens; 1 means it always does.

When every outcome in the sample space is **equally likely**,

$$P(A) = \frac{\text{number of ways } A \text{ can happen}}{\text{total number of outcomes}}$$

so for a fair die $P(\text{even}) = 3/6 = 0.5$ and $P(\text{not a six}) = 5/6$.

## Example: probabilities from real data

The table below classifies the 379 students in `heights.csv` by gender and by whether they are at least 70 inches tall. If we pick one student at random, every student is equally likely, so each probability is just a count divided by 379.

| Gender | At least 70 in | Under 70 in | Total |
|---|---|---|---|
| Female | 19 | 243 | 262 |

| Gender | At least 70 in | Under 70 in | Total |
|---|---|---|---|
| Male | 87 | 30 | 117 |
| Total | 106 | 273 | 379 |

Let $F$ be the event that the student is female and $T$ that the student is at least 70 inches tall. Then $P(T) = 106/379 \approx 0.28$ and $P(F) = 262/379 \approx 0.69$.

## Where do probabilities come from?

Flip a fair coin once. Intuition says $P(\text{heads}) = 1/2$. But suppose you flip it 4 times and get 3 heads, 75%. Was 1/2 wrong?

No. With random phenomena, the proportion of times an event happens can be quite unpredictable in the short run but very predictable in the long run.

That is exactly the definition of probability from the start of this section: the **probability** of an event is the proportion of times it occurs in a very long series of repetitions, not in a handful. The law of large numbers is what makes that long-run proportion settle down to a single number.

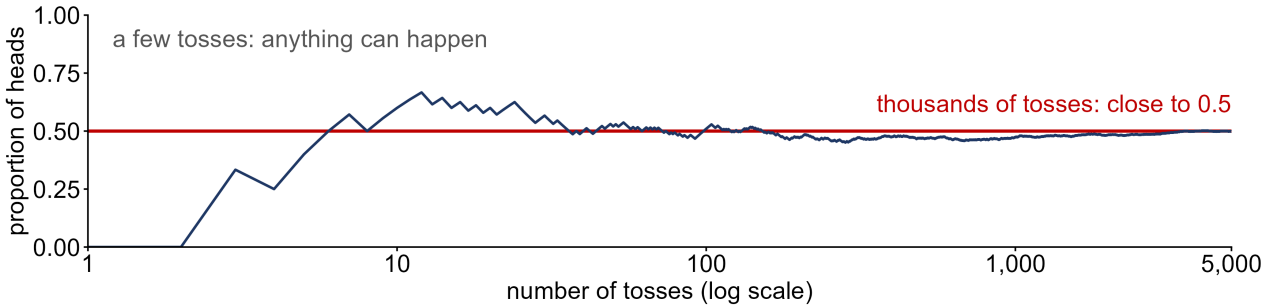
## The law of large numbers

- The **law of large numbers** (LLN): as the number of independent trials $n$ increases, the relative frequency of an event,

$$\frac{\text{number of times } A \text{ occurs}}{n},$$

  gets closer and closer to its probability $P(A)$.
- Trials are **independent trials** if the outcome of any one trial is not affected by the outcome of any other.

The plot below follows the running proportion of heads over 5,000 simulated tosses of a fair coin.



In the first few tosses the proportion jumps all over the place; over thousands of tosses it settles down to 0.5. The Swiss mathematician Jacob Bernoulli proved the first version of the law; it was published in 1713, after his death, in *Ars Conjectandi*. The statistician Karl Pearson checked it by hand, tossing a coin 24,000 times and getting 12,012 heads, a proportion of 0.5005.

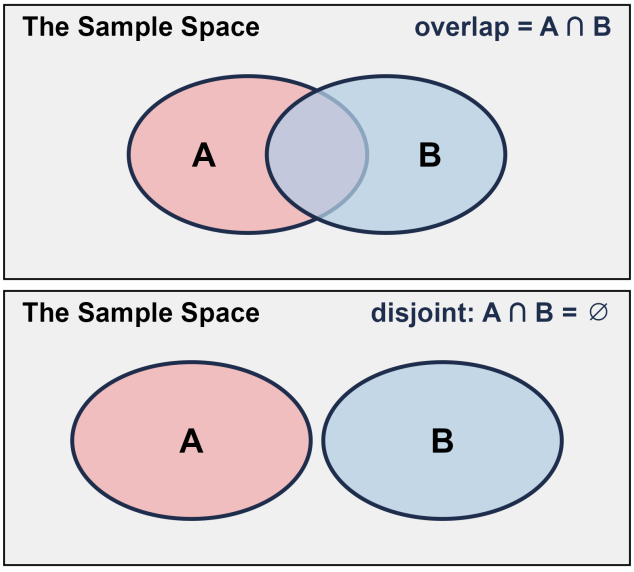
The LLN is also how probabilities are found when outcomes are not equally likely: the relative frequency over many trials is an *estimate* of the probability, and the estimate gets better as $n$ gets larger. That is exactly what the heights table did: $P(T) \approx 0.28$ is a relative frequency.

## The rules of probability

1. $0 \leq P(A) \leq 1$: the probability of an event must be a value between 0 and 1.
2. $\sum_i P(A_i) = 1$: the probabilities of all the outcomes in the sample space must add to 1.
3. $P(A \cup B) = P(A) + P(B)$ when $A$ and $B$ are **disjoint**, that is, they have no outcomes in common. This is the **addition rule for disjoint events**.
4. $P(A') = 1 - P(A)$: the **complement rule**. $A'$ ("$A$-not") is every outcome in $S$ that is not in $A$.

## Combining two events

- The **union** $A \cup B$ is all the outcomes in $A$ or in $B$ (or both). The key word is "or".
- The **intersection** $A \cap B$ is the outcomes that $A$ and $B$ have in common. The key word is "and".
- $A$ and $B$ are **disjoint** (or **mutually exclusive**) if they have no outcomes in common: $A \cap B = \emptyset$.



Two more rules for combining events:

- The **general addition rule**, for events that may overlap:

  $$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

  The subtraction stops the outcomes in the overlap from being counted twice. For disjoint events $P(A \cap B) = 0$, and this is just Rule 3.
- The **multiplication rule for independent events.** Two events are **independent** if one occurring does not change the probability of the other. Then

  $$P(A \cap B) = P(A) \cdot P(B)$$

- **Disjoint $\neq$ independent.** Disjoint events cannot happen together, so knowing one happened tells you the other did not, and that makes them strongly dependent.

## Example: one die

Roll a fair six-sided die once.

- $P(\text{not a six}) = 1 - 1/6 = 5/6$, by Rule 4.
- "Roll a 1" and "roll a 6" are disjoint, so $P(1 \cup 6) = 1/6 + 1/6 = 1/3$, by Rule 3.
- "Roll an even number" and "roll a number greater than 3" overlap (on a 4 or a 6), so by the general addition rule

  $$P(\text{even} \cup > 3) = \frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \frac{4}{6}$$

  Check it directly: the outcomes are $\{2, 4, 5, 6\}$, four of the six.

## Example: two dice

Roll the die twice. The two rolls are independent.

- $P(\text{six on both rolls}) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$, by the multiplication rule.
- $P(\text{at least one six})$ is easiest through the complement, since "at least one six" is the complement of "no sixes":

  $$P(\text{at least one six}) = 1 - P(\text{no sixes}) = 1 - \frac{5}{6} \cdot \frac{5}{6} = \frac{11}{36} \approx 0.31$$

## Example: unions and intersections

Back to the heights table, with $F$ = female and $T$ = at least 70 inches.

- $P(F \cap T) = 19/379 \approx 0.05$: female **and** at least 70 inches.
- By the general addition rule,

$$P(F \cup T) = \frac{262}{379} + \frac{106}{379} - \frac{19}{379} = \frac{349}{379} \approx 0.92$$

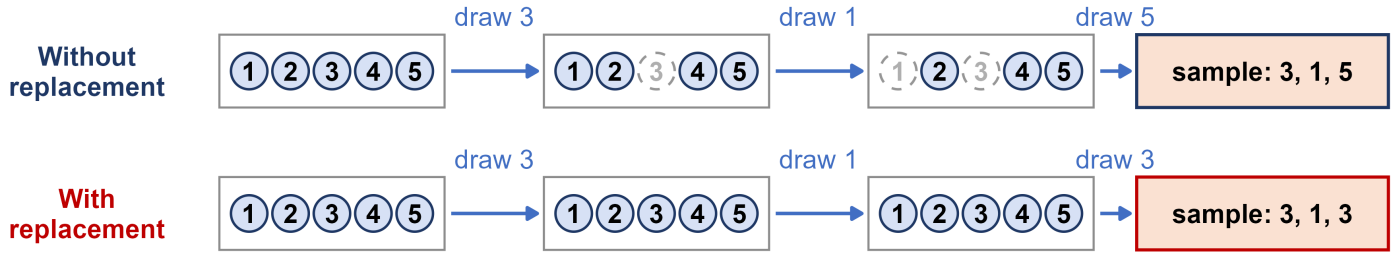
  Without the subtraction, the 19 tall women would be counted twice.

### Are $F$ and $T$ independent?

No. Among all students 28% are at least 70 inches, but among the women only $19/262 \approx 7\%$ are, so knowing a student is female changes the probability a lot. The multiplication rule would predict $P(F) \cdot P(T) \approx 0.69 \cdot 0.28 \approx 0.19$, far from the actual $P(F \cap T) \approx 0.05$.

## Sampling with and without replacement

Recall the two ways of drawing a sample from the Weeks 4-5 notes. Here a sample of 3 balls is drawn from a bag of 5:




Are the draws independent?

- **With replacement**, yes. Each ball goes back before the next draw, so the bag is the same on every draw and what happened on one draw cannot change the next. The multiplication rule $P(A \cap B) = P(A) \cdot P(B)$ applies.
- **Without replacement**, no. Each draw removes a ball, so it changes what is left, and the probabilities on the next draw depend on what was drawn before.

Simple random samples are drawn without replacement, so strictly their draws are dependent. When the population is much larger than the sample, though, removing a few units barely changes what is left, and the draws behave almost exactly as if they were independent.

### Example: drawing from an urn

An urn holds 10 balls: 3 red, 4 blue and 3 yellow. Draw 2 balls **without replacement**. After each draw there is one fewer ball, and one fewer of the color just drawn.

- $P(\text{both red}) = \frac{3}{10} \cdot \frac{2}{9} = \frac{6}{90} \approx 0.067$: after a red ball, only 2 of the remaining 9 are red.
- $P(\text{no red balls}) = \frac{7}{10} \cdot \frac{6}{9} = \frac{42}{90} \approx 0.467$: both draws come from the 7 balls that are not red.
- "Both the same color" means both red, both blue or both yellow. Those three events are disjoint, so add them:

$$P(\text{same color}) = \frac{3}{10} \cdot \frac{2}{9} + \frac{4}{10} \cdot \frac{3}{9} + \frac{3}{10} \cdot \frac{2}{9} = \frac{24}{90} \approx 0.267$$

### Example: drawing two aces

A standard deck has 52 cards, 4 of them aces. Draw two cards. What is the probability that both are aces?

- **With replacement:** the first card goes back, so the two draws are independent and

$$P(\text{two aces}) = \frac{4}{52} \cdot \frac{4}{52} = \frac{1}{169} \approx 0.0059$$

- **Without replacement:** if the first card is an ace, only 3 aces are left among 51 cards, so

$$P(\text{two aces}) = \frac{4}{52} \cdot \frac{3}{51} = \frac{1}{221} \approx 0.0045$$

Not putting the card back makes two aces less likely, because the first ace used up one of the four.

## Random variables

In statistics we usually record the outcome of a random trial as a number.

- A **random variable** is the outcome of a random trial recorded as a number, written with a capital letter such as $X$; a particular value is written with a lower-case letter such as $x$.
- It is **discrete** if its possible values can be listed, like a count of heads, and **continuous** if it can take any value in an interval, like a height. The normal distribution is the most important continuous distribution; this week we focus on discrete ones.

### Example: three coin tosses

Toss a fair coin 3 times. There are $2 \times 2 \times 2 = 8$ equally likely outcomes:

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

Let $X$ be the number of heads. HTH gives $X = 2$ and TTT gives $X = 0$, so $X$ can be 0, 1, 2 or 3.

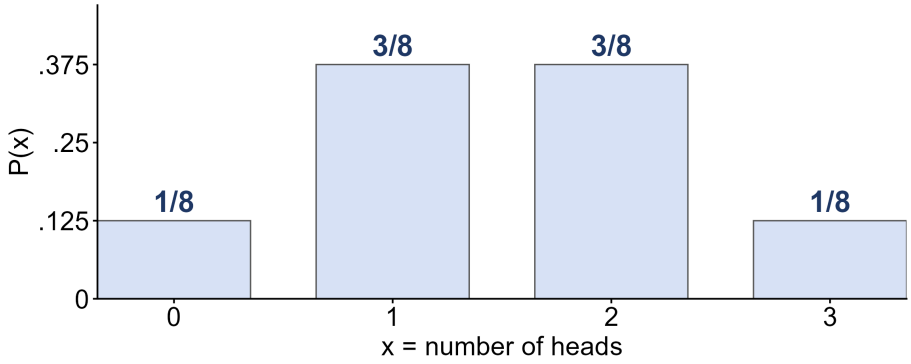
# Discrete probability distributions

- The **probability distribution** of a discrete random variable is a function that assigns a probability to each possible value of $X$. Each probability is between 0 and 1, and they add to 1. (The textbook calls this a *probability distribution function*, or PDF.)

For the three coin tosses, counting the outcomes that give each number of heads:

| $x$ (heads) | Outcomes | Number of outcomes | $P(x)$ |
|---|---|---|---|
| 0 | TTT | 1 | $1/8$ |
| 1 | HTT, THT, TTH | 3 | $3/8$ |
| 2 | HHT, HTH, THH | 3 | $3/8$ |
| 3 | HHH | 1 | $1/8$ |

Here $1+3+3+1=8$ outcomes, so the probabilities add to $8/8=1$.




To find the probability of an event, add the probabilities of the values that make it up:

- $P(X \geq 2) = P(X=2) + P(X=3) = 3/8 + 1/8 = 4/8$.
- $P(X \leq 1) = 1 - P(X \geq 2) = 1 - 4/8 = 4/8$, by the complement rule. In general $P(X > a) = 1 - P(X \leq a)$.
- $P(1 \leq X \leq 2) = 3/8 + 3/8 = 6/8$.

# Mean and standard deviation of a discrete random variable

A probability distribution has a center and a spread, just like a distribution of data. The formulas weight each value by its probability instead of by how often it appeared in a sample.

- The **mean**, or **expected value**, of a discrete random variable is its long-run average:

$$\mu = \sum x \cdot P(x)$$

- The **variance** and **standard deviation** are

$$\sigma^2 = \sum (x - \mu)^2 \cdot P(x) \qquad\qquad \sigma = \sqrt{\sigma^2}$$

The expected value is the long-run average of $X$ over many repetitions, which is the law of large numbers again. It does not have to be a value $X$ can actually take, and it is not a prediction for any single trial.

For the number of heads in three tosses:

| $x$ | $P(x)$ | $x \cdot P(x)$ | $(x-\mu)^2 \cdot P(x)$ |
|---|---|---|---|
| 0 | $1/8$ | $0(1/8) = 0.000$ | $(0 - 1.5)^2(1/8) = 0.281$ |
| 1 | $3/8$ | $1(3/8) = 0.375$ | $(1 - 1.5)^2(3/8) = 0.094$ |
| 2 | $3/8$ | $2(3/8) = 0.750$ | $(2 - 1.5)^2(3/8) = 0.094$ |
| 3 | $1/8$ | $3(1/8) = 0.375$ | $(3 - 1.5)^2(1/8) = 0.281$ |

Adding the columns, $\mu = 0 + 0.375 + 0.75 + 0.375 = 1.5$ heads, and $\sigma^2 = 0.281 + 0.094 + 0.094 + 0.281 = 0.75$, so $\sigma = \sqrt{0.75} \approx 0.87$ heads. Over many sets of three tosses the average number of heads is 1.5, even though no single set can have 1.5 heads.

## Example: grade points on an exam

A STAT 251 exam grade on the four-point scale ($F = 0$, $D = 1$, $C = 2$, $B = 3$, $A = 4$) is a random variable $X$ for a student chosen at random. Suppose its distribution is:

| $x$ | Grade | $P(x)$ |
|---|---|---|
| 0 | F | 0.05 |
| 1 | D | 0.10 |
| 2 | C | 0.30 |
| 3 | B | 0.25 |
| 4 | A | 0.30 |

- The probabilities add to $0.05 + 0.10 + 0.30 + 0.25 + 0.30 = 1$, so this is a valid distribution.

- The probability of doing better than a C is $P(X > 2) = P(X = 3) + P(X = 4) = 0.25 + 0.30 = 0.55$, or through the complement, $1 - (0.05 + 0.10 + 0.30) = 0.55$.
- The mean grade is $\mu = 0(0.05) + 1(0.10) + 2(0.30) + 3(0.25) + 4(0.30) = 2.65$, between a C and a B.
- The standard deviation is $\sigma = \sqrt{\sum (x - 2.65)^2 P(x)} = 1.15$ grade points.

Next week we meet some named discrete distributions that come up again and again, starting with the Bernoulli and the binomial, and then return to the sampling distribution with the central limit theorem.

# Conclusion: what you should take away

## The normal distribution and $z$-scores

- $X \sim N(\mu, \sigma)$: the mean locates the center and the standard deviation sets the spread.
- **The empirical rule:** about 68%, 95% and 99.7% of values within 1, 2 and 3 standard deviations of the mean, for bell-shaped distributions only. With symmetry it also gives 84%, 16%, 97.5% and 2.5%.
- **The $z$-score** $z = (x - \bar{x})/s$ counts standard deviations from the mean. It has no units, which is what lets you compare values measured on different scales.
- **Outliers:** $|z| \geq 2$ is the usual flag for bell-shaped data. It can disagree with the $1.5 \times IQR$ rule, because one extreme value inflates $s$ but barely moves the quartiles.
- For $z$-scores between whole numbers, get the area from a $z$-table or from `pnorm` / `qnorm`.
- A linear transformation $y = ax + b$ moves the center, multiplies the spread by $|a|$ and never changes the shape, so a $z$-score does not make skewed data normal.

## Sampling distributions

- A statistic varies from sample to sample. Its **sampling distribution** is its distribution over all possible samples of size $n$.
- A sampling distribution is centered at the parameter only if the statistic is **unbiased**. For a random sample, $\bar{x}$ is unbiased, so its sampling distribution is centered at $\mu$ and has **standard error** $SE = \sigma/\sqrt{n}$. Quadrupling $n$ halves the standard error.
- **Margin of error** $= 2 \cdot SE$. A result more than 2 standard errors from what chance alone would give is **statistically significant**, which is not the same as important.

## Probability

- Probability is the long-run relative frequency of an event, between 0 and 1. The **law of large numbers** says the relative frequency over $n$ independent trials gets closer and closer to the probability as $n$ grows.
- The four rules: $0 \le P(A) \le 1$; $\sum_i P(A_i) = 1$; $P(A \cup B) = P(A) + P(B)$ for disjoint events; $P(A') = 1 - P(A)$. In general $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, and for independent events $P(A \cap B) = P(A) \cdot P(B)$. Disjoint is not the same as independent.
- A **random variable** is the outcome of a random trial recorded as a number. A discrete one has a probability distribution with mean $\mu = \sum x P(x)$ and standard deviation $\sigma = \sqrt{\sum (x - \mu)^2 P(x)}$.

## Sources


- Illowsky, B. and Dean, S. (2023). *Introductory Statistics 2e*, Sections 3.1–3.3, 4.1–4.2, 6.1–6.2 and 7.1. OpenStax. OpenStax: Introductory Statistics 2e (https://openstax.org/details/books/introductory-statistics-2e)
- World Bank. CO2 emissions (metric tons per capita), indicator EN.ATM.CO2E.PC, for the 28 member states of the European Union. World Bank: CO2 emissions (metric tons per capita) (https://data.worldbank.org/indicator/EN.ATM.CO2E.PC)
- National Weather Service, New York, NY. Central Park monthly mean temperatures, 1869–2018 (the January column of `Data/central_park_temps.csv`).
- Pearson's 24,000 coin tosses are reported in Illowsky and Dean (2023), Section 4.2.